

DAS COMPETITION
PROTOTYPING & MVPs



WHY PROTOTYPE?

Prototyping allows you to iteratively learn and improve your product

The goal is to collect the maximum amount of validated learning with the least time, cost & effort..

Ultimately the goal is **product-market fit**.

This is done by building and releasing a **Minimum Viable Product**

A SIMPLE PROCESS
BUILD – MEASURE – LEARN

Build a prototype or MVP

›

Let users use it

›

Get user feedback early and often

›

Fail fast and cheaply

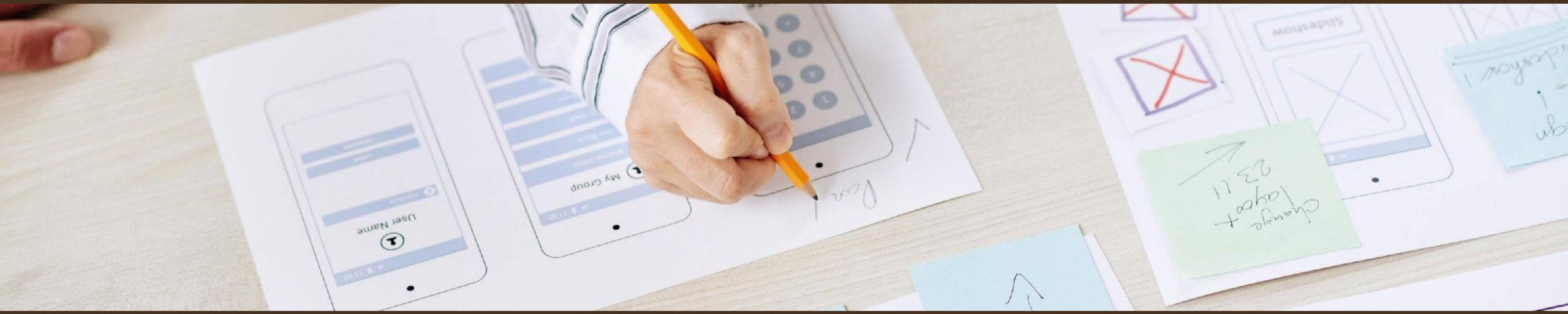
›

Iterate and improve

›



PROTOTYPING IS
ITERATIVE



DIFFERENT PROTOTYPES
FOR DIFFERENT PURPOSES

THE PROTOTYPING PROCESS

LOW FIDELITY
less time
lower cost

HIGH FIDELITY
more time
higher cost



THE PROTOTYPING PROCESS

LOW FIDELITY
less time
lower cost

HIGH FIDELITY
more time
higher cost

IDEA

LOOKS LIKE

WORKS LIKE

INTEGRATED PRODUCT



THE PROTOTYPING PROCESS

LOW FIDELITY
less time
lower cost

HIGH FIDELITY
more time
higher cost

IDEA → **LOOKS LIKE** → **WORKS LIKE** → **INTEGRATED PRODUCT**

RESOLUTION, MATERIALS, FUNCTIONALITY, QUALITY, SCALE

sketches, cardboard, foamcore, wood,
clay, paper mockups, no-code

CAD, 3D prints, circuits, code
low volume manufacturing



EVERY PROTOTYPE HAS A **PURPOSE & COST**

COMMUNICATE
IDEAS

- Sketches
- Storyboards
- Cardboard Mockups
- Explainer Videos

USER
EXPERIENCE

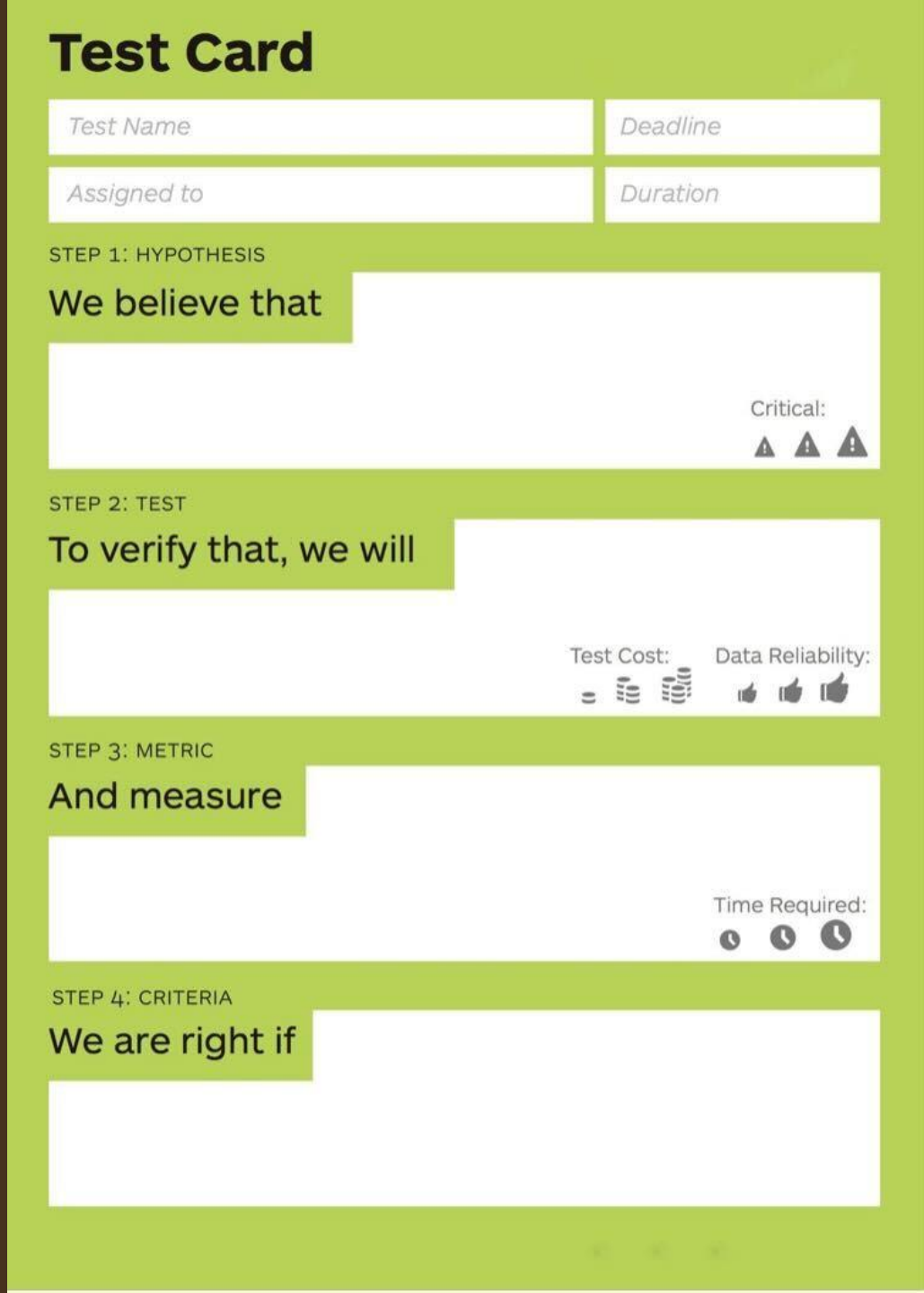
- Paper Mockups
- Foam Mockups
- Wireframes
- 3D Prints
- Clickable Mockups
- Appearance Models
- Looks-like Works-like

TECHNICAL
FUNCTIONALITY

- Proof of Concept
- Breadboard mockups
- Digital simulations
- Works-Like prototypes



KNOW WHAT YOU ARE TESTING AND WHAT YOU HOPE TO **LEARN**



Testing Business Ideas
by David J Bland and Alex Osterwalder

PROTOTYPES VS MVPS

PROTOTYPE
(not an MVP)

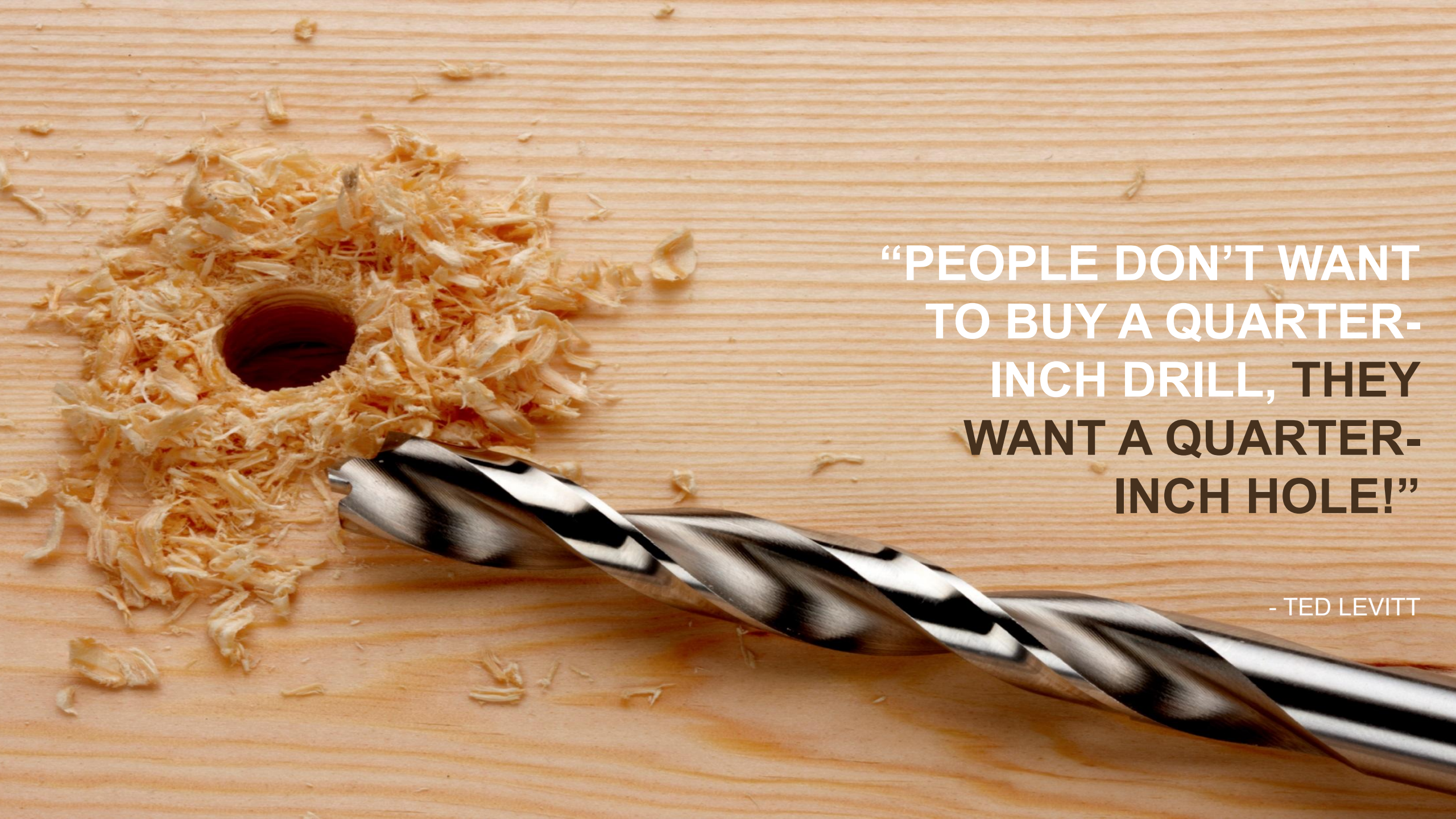


MVP



MINIMUM VIABLE PRODUCT

- The goal of an MVP is to learn as quickly and cheaply as possible whether there is a market for your product.
- “We must learn what customers really want, not what they say they want or what we think they should want.”
- “Remove any feature, process, or effort that does not


“PEOPLE DON’T WANT TO BUY A QUARTER-INCH DRILL, THEY WANT A QUARTER-INCH HOLE!”
- TED LEVITT



PRODUCT-MARKET FIT

- Solving an important problem
- Better than the alternatives
- For a defined market
- With customers willing to pay

Products that don't achieve product market fit eventually fail

HOW DO WE KNOW WE HAVE PRODUCT **MARKET FIT?**



Customers show a willingness to pay



They are using the product regularly (and continue using it)



They are recommending it to others

WHAT MAKES A **MINIMUM VIABLE PRODUCT**

MINIMUM

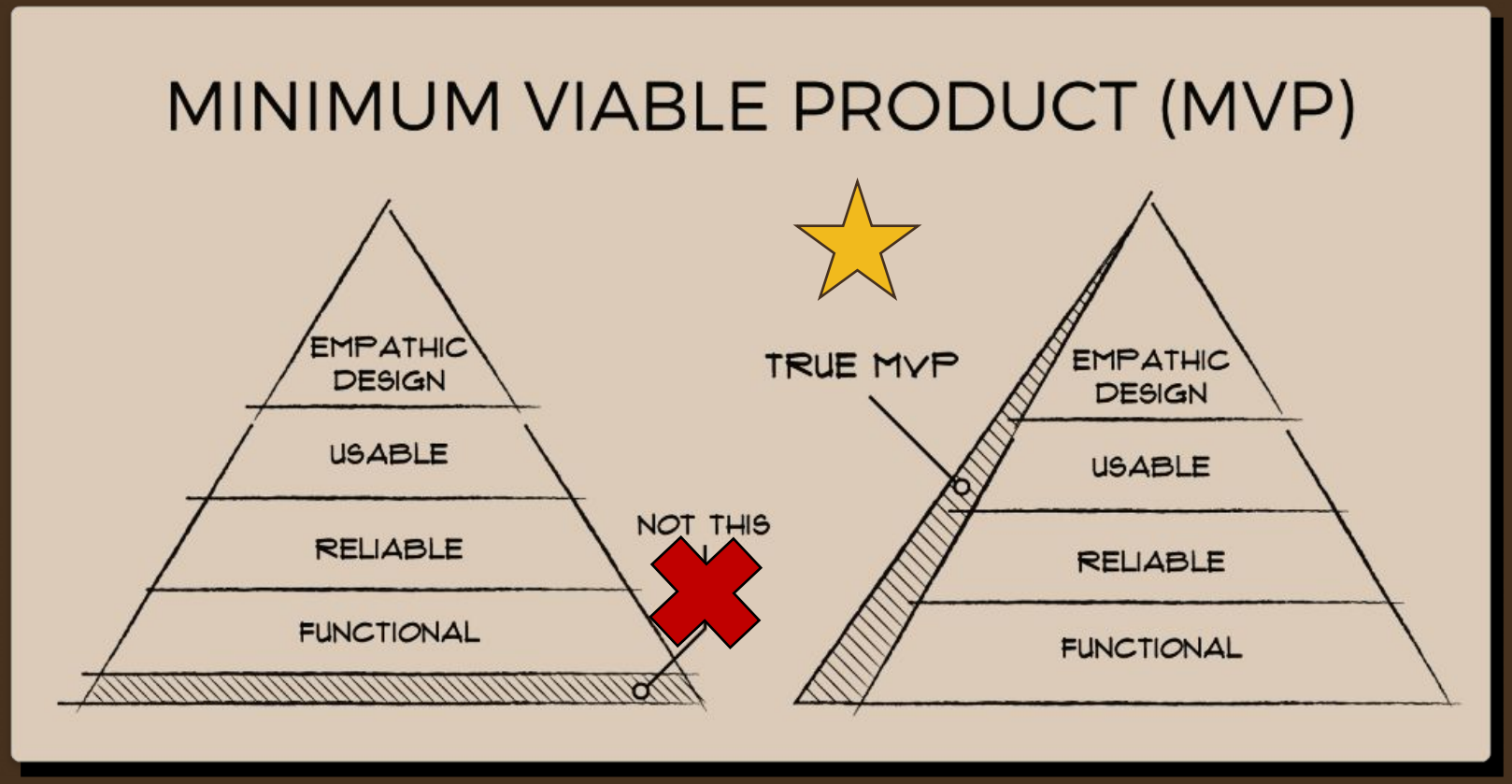
The product should have the **bare minimum features necessary to solve the user's problem.** This means avoiding adding any features that are not essential to the product's core functionality.

VIABLE

The product should be able to **attract early adopters** and **generate meaningful feedback** to help the product team iterate and improve the product over time. This means that the product should be **functional and user-friendly**, even if it is not perfect.

If we can’t viably solve early customers’ primary problems, nothing else matters.

MVP STRUCTURE



An MVP needs some basic design and user experience considerations too.

Good design
- helps attract users
- helps you get better user feedback
- builds trust with users.

MVP FEATURE SELECTION

MUST HAVES

Any product without these features will not be considered by users

INCLUDE ALL

DIFFERENTIATORS

Features that differentiate your product and make it better than the alternatives

INCLUDE **1-3 MAXIMUM**

NICE TO HAVE

Nice but non-essential features that would enhance the product or may add value to the user

IGNORE FOR NOW

MVP Experiment Canvas

Project Name:	Made by:	START date / time: END date / time:

4. Customer Engagement ❤️

How are you going to engage the customers who used your MVP, to learn from their experience and gather feedback to improve your Value Proposition?

1. Your Customer Segment 👤

Which segment of your target customers will experience the most value from your MVP, and can you learn from them?

2. Value Proposition 🎁

My company, __(insert of company)____,
is developing __(a defined offering)__
to help __(a defined audience)__
__(solve a problem)__
with __(secret sauce/differentiation)__.

3. Channel(s) 🚚

How are you going to reach your Customer Segment and “deliver” your Value Proposition to them?

7. Experiment Scenario / Workflow 🔄

HOW does that work? Describe the steps from beginning to end.

8. Metrics 📈

What will you measure? (Can be multiple metrics)

10. Results 🏁

What are the qualitative and / or quantitative results of your experiment?

11. Learnings & Insights 🔑

What are your key learnings?

What are your key surprises?

Did you get enough results?

What do they tell you about your Riskiest Assumption(s)?

VALIDATED	INVALIDATED	INCONCLUSIVE

5. Riskiest Assumption(s) 💣

What you are setting out to test with your MVP?

Are these assumptions about the Problem, Solution or Implementation?

6. Experiment Format ⚗️

WHAT type of experiment are you going to run?

9. Success Criteria 🏆

How do you qualify and/or quantify a successful outcome of this experiment?

12. Next steps 🚦

What do the results tell you to do next?

Are you going to Pivot, Pursue, or Stop?

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Learn more and download the canvas at **themvpcanvas.com**

Made by **Bram Kanstein (twitter.com/bramk)**

IN SUMMARY

1. The process: build – test – iterate…. over and over and over
2. Build fast- do not build a full product, do not worry about scaling or monetizing
3. Prepare for it to take 10x more iterations than expected. Keep costs low!
4. Know who is your target customer and focus on them
5. Testing: customer interviews; observation with product (no helping)

ONLY ONE THING MATTERS:
what do you need to learn NOW to get to the next step?
OPTIMIZE PROTOTYPES FOR LEARNING



TYPES OF MVPS

1: LANDING **PAGE**

This is a simple website that describes the product and allows users to sign up for a waitlist or newsletter.

EXAMPLE

Buffer - Before developing their social media management tool, the Buffer team created a landing page describing the product's features and benefits. They then monitored the interest and sign-ups to gauge demand and decide whether to proceed with development

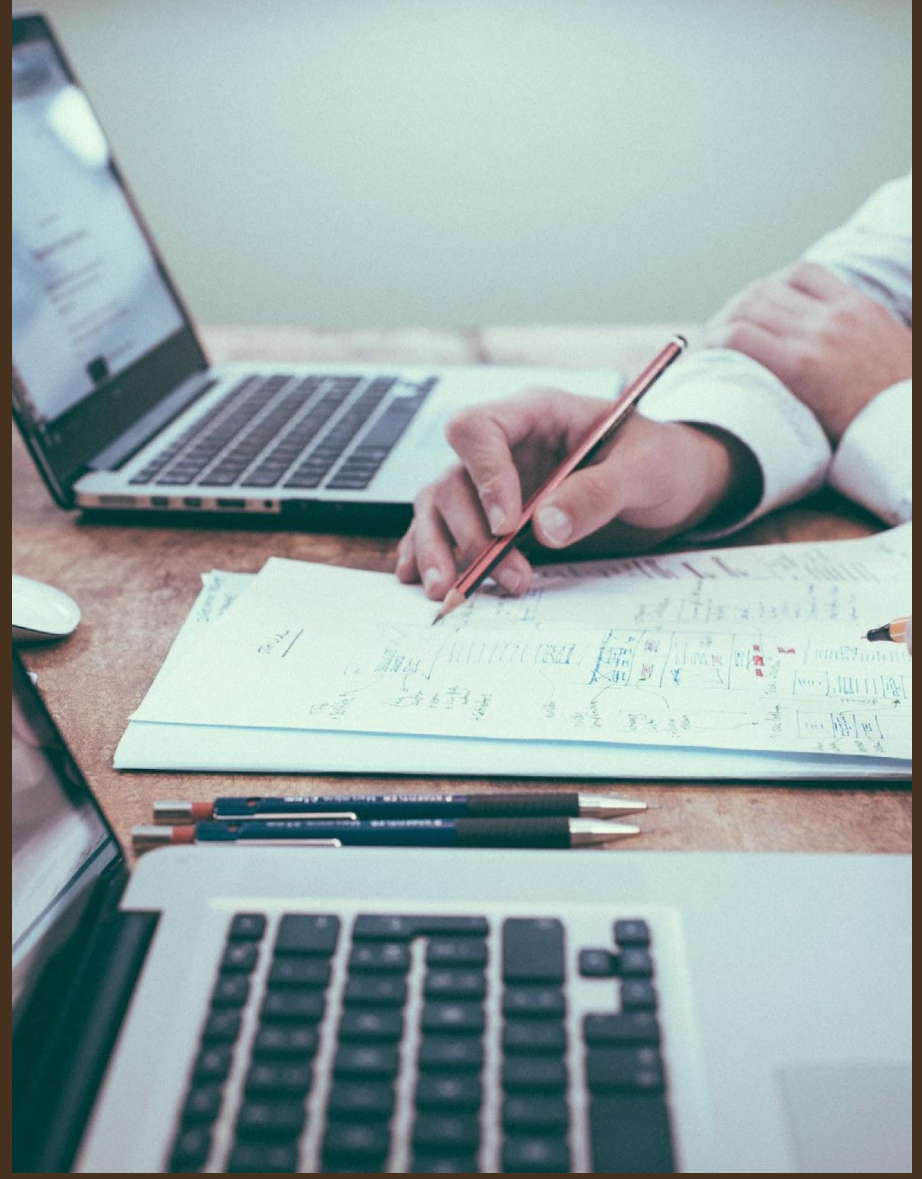


2: EMAIL CAMPAIGN

This is an email campaign that introduces the product and collects feedback from users. Often the campaign collects emails of interested customers.

EXAMPLE

Product Hunt used an email MVP to test the interest in their product discovery platform.





3: WIZARD OF OZ

This is a product that appears to be fully automated, but it is actually being manually operated by a team of people.

EXAMPLE

Airbnb - In the early days, Airbnb founders Brian Chesky and Joe Gebbia set up a simple website to offer air mattresses in their living room to attendees of a design conference in San Francisco. They built a simple page where they posted pictures of their apartment, the price per night, contacts, and address.

They handled bookings and interactions manually to test the concept before automating the process.

4: SINGLE FEATURE

EXAMPLE

- **Dropbox** - The initial MVP for Dropbox was a simple 3 minute video demonstrating the core functionality of syncing files to the cloud and accessing them from any device.

- The film was published on the Hacker News website - an integrator of news from social media, devoted to topics related to startups and technology, i.e. where potential first adopters of the service look.

- This single feature was enough to gauge interest and attract early adopters.



4: EXPLAINER VIDEO

EXAMPLE

- **Dollar Shave Club** - Instead of creating a full-scale shaving product and infrastructure, Dollar Shave Club started with a humorous explainer video showcasing their subscription service.

- There was no actual product yet.

- The video went viral, and they received enough sign-ups to validate the market before actually manufacturing the razors.



5: CONCIERGE SERVICE

EXAMPLE

Zappos - Before building a full-scale e-commerce website, Zappos founder, Nick Swinmurn, started by taking pictures of shoes in local shoe stores and posting them online.

When customers made a purchase, he would buy the shoes from the store and ship them to the buyers, essentially validating the demand for an online shoe store without building the infrastructure upfront.

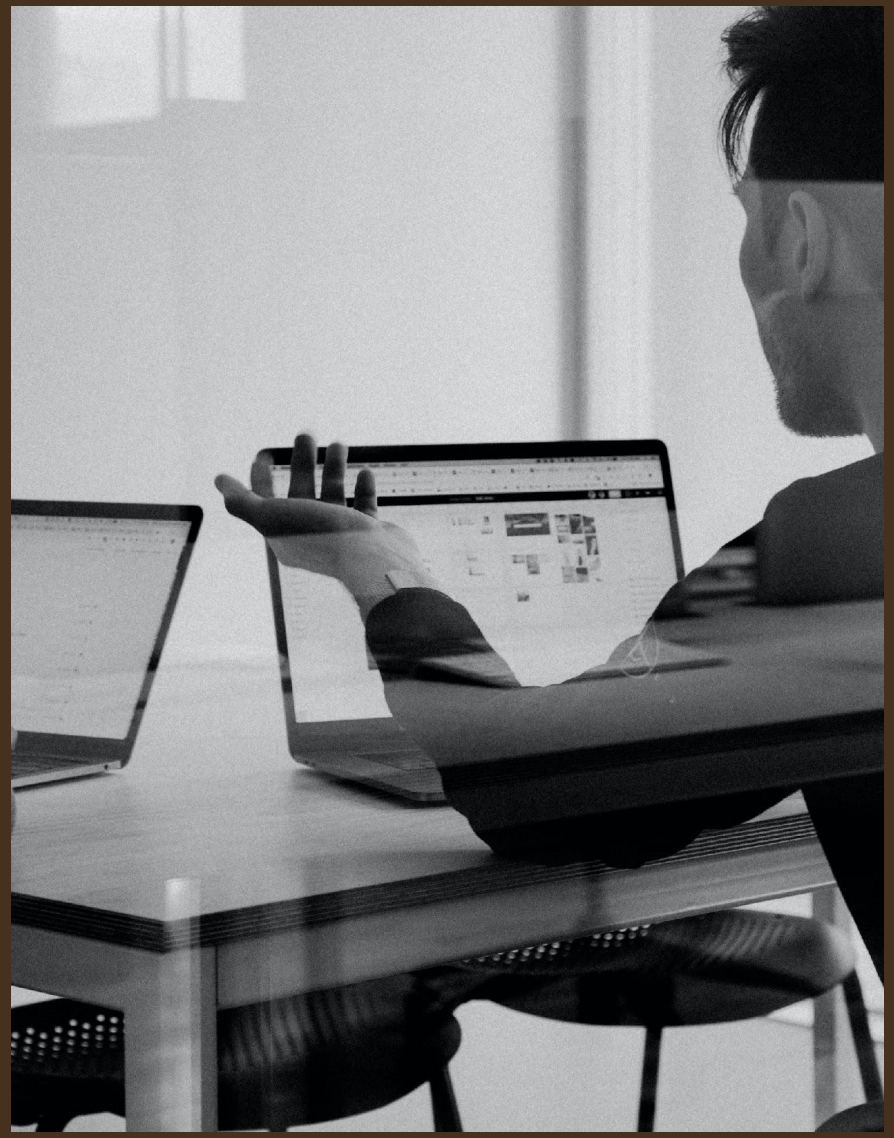


5: WIREFRAME/MOCKUP

EXAMPLE

- **Uber** - Before writing any code, Uber's founders created a basic wireframe of the app to demonstrate how it would work. They then tested the concept with friends and family to validate the idea.

- When they did finally build an app, the first version had very simple graphics, was in one city only, and you could not pay through the app.





5: PRE-ORDER

EXAMPLE

Oculus Rift - Oculus, a virtual reality company, launched a Kickstarter campaign to fund the development of their VR headset.

The overwhelming response from backers validated the demand for the product and helped secure additional funding.



DAS COMPETITION
RESOURCES

EXAMPLES OF VARIOUS PHYSICAL **PROTOTYPES**

TYPE	PURPOSE
Sketch / Concept Art	initial brainstorming, ideation or visualization
Lo-Fidelity mock up	quick mockups to evaluate use cases + ideas in reality
Breadboard mockup	basic working model of idea or function
Aesthetic model	Looks exactly like final product but no functionality
Functional model	Functional but not final design or parts
Proof of Concept	Implementation of a certain method/idea using specific
Scale model	Used for objects too large to easily prototype full size
Looks-like-works-like	May not be using final production parts
Ergonomic model	used for human interaction testing
MVP	simplest prototype acceptable to test product-market fit

EXAMPLES OF VARIOUS DIGITAL **PROTOTYPES**

TYPE	PURPOSE
Storyboard	Tell the story in frames
Sketch / Concept Art	initial brainstorming, ideation or visualization
Digital Wireframe	Skeleton framework that is used as a basis for design
Screen mockups	mockup of UI screens (can be paper drawings, or digital)
Static prototype	appears to be functional but is hardcoded
Clickable mockups	fake simulation of software UI/UX
Horizontal proto	Broad Front-end (i.e. User interface) of an entire system
Vertical proto	Deep prototype of a complete function (includes Back–end)
Demo Video	Used to simulate or illustrate product, use case or environment
MVP	simplest prototype acceptable to test product-market fit

BUILDING AN MVP



- Minimize the costs required for product testing
- Obtain feedback from users for making timely changes
- Reduce the time spent on development
- Ensure fast product release for early users
- Attract investors
- Detect and prevent future pitfalls

Ideation

Value proposition mapping

Overview

- Map the value proposition of your solution to the needs and wants of your key stakeholders (users, customers)

30 minutes

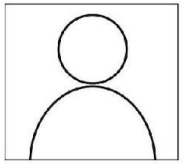
Paper

Instructions

It is important that your solution meets the needs and wants of your key stakeholders, including users (who will use the solution), customers (who will pay for the solution; can be the same profile as users), funders (who will provide capital to launch the solution), and partners (who will help execute the solution). See *Testing: Who to Speak to* for more details on the four above-mentioned stakeholder groups.

1. On a large sheet of paper, write the name of the stakeholder group and the name of the specific person being profiled at the top.

Stakeholder Group: Name



Name, age, gender, job, etc.

	Stakeholder	Solution
Needs	*What the stakeholder requires*	*Value proposition of solution to address needs*
Wants	*What the stakeholder desires*	*Value proposition of solution to address wants*

2. Outline the stakeholder's needs and wants. Note that these are not necessarily the same thing. **Needs** are what the stakeholder requires, while **wants** are what the stakeholder desires.
3. Outline your solution's value proposition, or how your solution addresses both the stakeholder's needs and wants.
4. Repeat for all stakeholder groups.

Ideation

PATH Statement

Overview

- Revisit your problem framing to include your solution
- Succinctly define your problem, approach, target, and heart & humanity

30 minutes

Paper

Instructions

A PATH statement is a succinct statement that addresses four elements:

- **Problem**: What problem are you solving?
- **Approach**: How are you solving this problem?
- **Target**: Who are the users and customers that you are reaching with your solution?
- **Heart & Humanity**: What is the impact you are trying to create with this solution and why do you care about creating this impact?

A PATH statement builds upon your problem framing, adding the elements of the proposed solution. For solutions that address global challenges, it is important to include not just the impact that your solution is making, but also your personal passion for solving this challenge, which will help keep you motivated in the face of adversity.

A PATH statement can be a good starting point for a 60 second pitch on your problem and proposed solution.

As you continue through the design process, it is important to re-visit your PATH statement and ensure you are staying on Target and staying true to the Heart & Humanity of the solution.

Prototyping

Design specifications

Overview

- Outline your assumptions
- Turn your assumptions into tangible design specifications that can be used for guiding how to make your prototypes

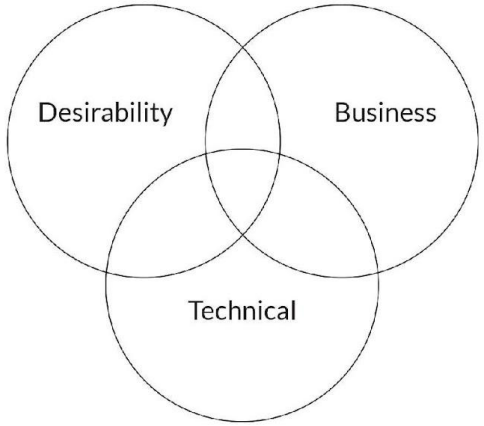
1 hour

Paper

Instructions

Assumptions are things that you assume about your problem, customer, or solution, which need to be verified on a large scale.

Design specifications are the quantitative and qualitative specifications that guide your solution. They fall into three categories: desirability, business, and technical.



Desirability specifications pertain to user wants and needs (e.g. the look and feel of the solution, or the want/need it fills). **Business** specifications pertain to the operations of delivering this solution (e.g. economic constraints or delivery models). **Technical** specifications pertain to the functionality of the solution itself (e.g. size, outputs, manufacturing - for a product; duration, content - for a service).

Specifications need to be measurable in order to make sure that you are meeting them. Thus, the more you can quantify the above specifications and constraints, the easier it will be to design all the details of your solution. User preferences are hard to quantify, but it is still helpful to try.

List all of your assumptions, and then list any corresponding design specifications that you have determined as a result of these assumptions.

Assumptions	Design specifications
List your assumptions here →	Quantify in terms of design specification.
For example:	**For example:**
Customers are extremely cost conscious →	Manufacture for less than **$10**
Assembly must be quick and easy →	Can be assembled by **one person** in less than **8 minutes**
Needs to be transported on the back of a motorbike →	Can be packed into a box less than **50x50x50 cm** and weighs less than **20 kg**
New sales agents must start immediately after hiring →	Can be trained and onboarded in less than **15 minutes**
Customers need to like the product (user preferences) →	At least **80%** of surveyed users select new product vs. the competition

Prototyping

Rapid prototyping

Overview

- Quickly create a physical representation of your solution that a user can experience

1-3 hours

Cardboard, foam core, glue, clay, LEGO, fabric, wood, code, powerpoint

Instructions

We create prototypes so that we can get feedback on our ideas from users. Rapid prototyping is the act of quickly and cheaply creating a model of the solution, **which users can experience and react to**.

What to prototype

How much of the solution should you prototype? You can prototype an entire solution or a single aspect of the solution. Which one you choose depends on what you want to test during the Testing phase. If there are particular aspects of the solution that need testing, verification, or probing, the prototypes should be built with these in mind. Focus on these aspects and save time on the other aspects.

What if your solution is not a product? Anything is possible when creating prototypes. The only critical design criteria for a prototype are that it can be used to test one or more assumptions / questions. You can use storyboards, diagrams, narratives (tell a story), re-enactments, sketches (mock-ups), 3D models, or combine multiple elements, as long as you create testable features to answer your questions and test your assumptions.

How many prototypes can you build? As many as you can and need! This depends on what aspects of your solution need testing.

Types of prototypes

- **Works-like:** Prototypes that function like you want your solution to function.
- **Looks-like:** Prototypes that aesthetically look like how you want your solution to look.
- **Made-like:** Prototypes that are manufactured or created how you will ultimately manufacture your solution.

How to prototype

You can use a variety of tools at your disposal to create prototypes. Common prototyping materials include: paper, tape, scissors, cardboard, glue, foam core, wood, sheet metal, rivets, fabric, LEGO, clay, and more.

You can also create more detailed prototypes of physical solutions with software like Autodesk® Fusion 360™ (See *Prototyping: 3D Modeling with Autodesk®*). You can prototype a program with storyboards or a rough powerpoint animation.

Here are a few tips to get you started:

- **Start building.** It's ok if you don't know exactly how you're going to build the prototype yet. Just pick up some materials and experiment to help get the creative energy flowing.
- **Move fast**. Don't spend too long on any one prototype.
- **Identify a purpose for each prototype**. Note down what specific aspects you hope to test with this prototype, and who you will need to show it to in order to get the feedback that you need.

Prototyping

Priority vs. confidence mapping

Overview

- Map your assumptions and questions so you can identify which ones to verify or answer first

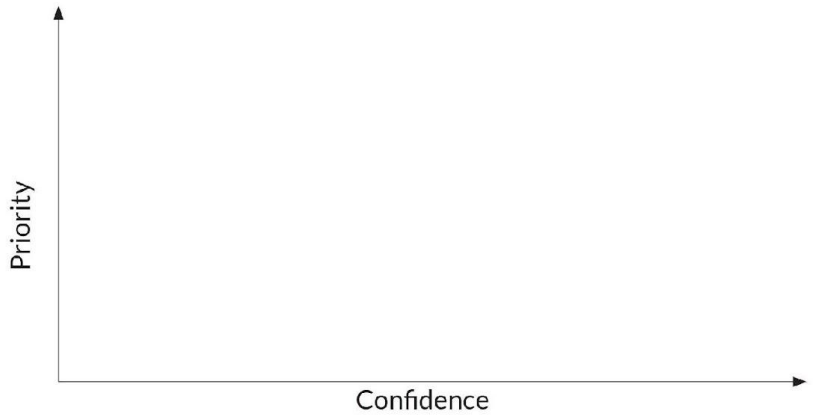
45 minutes

Paper, post-its

Instructions

To prioritize which assumptions to test and which questions to answer in the Testing phase, use a priority vs. confidence map.

1. Write all of your assumptions and questions on post-it notes. 1 assumption / question per post-it note.
2. Create a graph with "priority" on the y-axis and "confidence" on the x-axis.



3. Place each assumption and question on the graph based on how important they are to your solution, and your confidence in the accuracy of your information.
4. Focus on verifying assumptions and answering questions that are high priority, low confidence. The goal is to shift as many of these to high priority, high confidence, or to re-design your solution so your greatest uncertainties are lower priorities

